

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 2 (Generated practice paper - Set 32)**

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
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9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

0 1

Table 1 shows the numbers of breeding pairs of two seabird species on an island before and after a rat eradication programme.

Table 1

Species	Breeding pairs before eradication	Breeding pairs 8 years after eradication
Manx shearwater	45	310
Storm petrel	12	140

0 1

. 1

Explain why oceanic islands have many endemic species but low overall biodiversity.

[4 marks]

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. 2

Use Table 1 to explain why introduced rats had such a severe effect on these birds.

[3 marks]

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0 1

. 3

Explain how avian malaria threatens Hawaiian birds and how climate change worsens the threat.

[2 marks]

0 1 . 4

Give **one** other conservation measure for oceanic islands besides eradicating introduced species.

[1 mark]

10

0 2

Table 1 shows some of the IUCN criteria used to place species in Red List categories.

Table 1

Category	Population decline over 10 years or 3 generations	Area of habitat	Number of mature adults	Probability of extinction in the wild
Vulnerable	over 40%	under 2000 km ²	under 10 000	over 10% in 100 years
Endangered	over 60%	under 500 km ²	under 2500	over 20% in 20 years or 5 generations
Critically endangered	over 80%	under 10 km ²	under 250	over 50% in 10 years or 3 generations

0 2 . 1

State **three** roles of the IUCN.

[3 marks]

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2

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0 2 . 2

A species of lemur has declined from 12 000 to 2 100 mature adults in three generations and occupies 400 km² of forest. Use Table 1 to determine its Red List category. Explain your answer.

[3 marks]

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0 2 . 3

Explain why only a small proportion of species have been assessed for the Red List, and give **two** criteria used to decide which species to assess.

[3 marks]

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0 2 . 4

Give **one** example of a species whose Red List category improved because of conservation and state the actions responsible.

[1 mark]

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10

03 . 1

Explain what is meant by **biomimetics** and describe **three** examples.**[4 marks]**

03 . 2

Explain why many plants produce chemicals that are useful as human medicines, and give **two** examples.**[3 marks]**

03 . 3

Explain why the discovery of medicines is a strong argument for conserving whole habitats rather than individual species.

[2 marks]

03 . 4

Give **one** example of a wildlife species used in physiological research and the human health issue it helps to understand.**[1 mark]**

10

0 4 . 1

Give **two** natural processes that are driven by solar energy.

[2 marks]

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2

0 4 . 2

Explain why natural processes can take place at low temperatures.

[2 marks]

0 4 . 3

Give **one** human activity that uses high temperatures to achieve a result that natural systems achieve at low temperature.

[1 mark]

5

0 5 . 1

Explain the difference between primary and secondary succession and why secondary succession is faster.

[3 marks]

0 5 . 2

Explain what is meant by **deflected succession** and a **plagioclimax**, using a named UK example.

[3 marks]

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0 5 . 3

Explain why conservation of a plagioclimax requires continued human activity whereas conservation of a rainforest or coral reef does not.

[2 marks]

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0 5 . 4

Explain why biodiversity tends to be low in abiotically extreme environments and high in stable ones.

[2 marks]

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06 . 1

Explain why electronic meters must be calibrated before use and describe how a pH meter is calibrated.

[3 marks]

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06 . 2

Describe how wind velocity measurements with an anemometer should be standardised.

[3 marks]

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06 . 3

Describe how a whirling hygrometer is used to measure humidity.

[2 marks]

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06 . 4

Describe **two** methods of measuring nitrate concentration in water.

[2 marks]

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10

07.1 Describe the main principles of the circular economy.

[6 marks]

07 . 2 Explain what is meant by 'optimum production rather than maximum production', using a comparison between a natural ecosystem and an agroecosystem.

[2 marks]

0 7 . 3

Explain why international cooperation (eg through the EU or UN) is needed to implement circular economy initiatives.

[2 marks]

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10**0 8**

Table 1 shows some features of Earth, Mars and Venus.

Table 1

Feature	Venus	Earth	Mars
Mean distance from Sun / million km	108	150	228
Mass relative to Earth	0.82	1.0	0.11
Atmospheric pressure relative to Earth	92	1.0	0.006
Mean surface temperature / °C	460	15	-60
Magnetic field	Very weak	Strong	Very weak
Rotation period / hours	5 800	24	25

0 8 . 1

Use Table 1 to explain why liquid water cannot exist on the surface of Mars.

[3 marks]

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0 8 . 2

Explain how the Earth's mass and its distance from the Sun created suitable conditions for life.

[3 marks]

08 . 3

Explain the importance of the Earth's magnetic field and its speed of rotation for living organisms.

[2 marks]

08 . 4

Explain how the tilt of the Earth's axis affects living organisms.

[2 marks]

10

09 . 1

State the Brundtland Commission (1987) definition of sustainable development.

[1 mark]

09 . 2

Explain why activities such as slash-and-burn agriculture, fishing and sewage discharge can be sustainable with a small human population but not with a large one.

[3 marks]

09 . 3

Explain why urban populations are often less aware of unsustainable exploitation than rural communities.

[2 marks]

09 . 4

Describe **two** negative feedback mechanisms in natural systems and explain how human responses can override them.

[4 marks]

10

Discuss the strategies that can be used to increase the sustainability of agriculture.

[9 marks]

Do not write outside the box

Extra space

10 . 2 State what is meant by **permaculture**.

[1 mark]

10

Write an essay on one of the following topics.

$$\boxed{1} \boxed{1} . \boxed{1}$$

Fishing is the last large-scale human hunting activity. Discuss the environmental impacts of fishing and evaluate the methods used to make the exploitation of fish populations sustainable.

[25 marks]

OR

1 1 . 2

Discuss the extent to which the resources and services provided by other species justify the conservation of biodiversity.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

Do not write outside the box

A large rectangular box containing 30 horizontal dotted lines for writing.

END OF QUESTIONS